课堂作业二 考试重点

1. UNIT1 Software & Software Engineering（第1章）

1.1 How does software differ from the artifacts produced by other engineering disciplines?

中文翻译： 软件与其他工程学科产出的制品有何不同?

How does software differ from the artifacts produced by other engineering disciplines ?

Software is both a **product** and a **vehicle** for delivering a product.

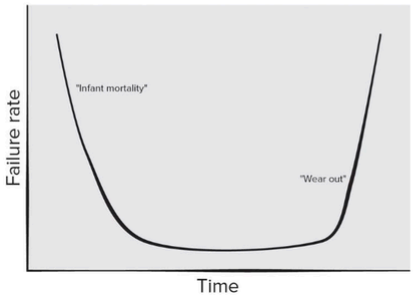
- As a **product**, software is an information transformer.
- As a **vehicle** for delivering a product, software serves as a basis for computer control, communication, and creation of other programs.

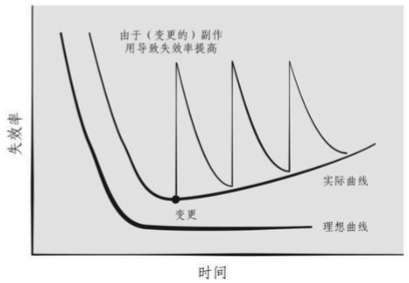
1.2 How do software characteristics differ from hardware characteristics?

中文翻译： 软件特性与硬件特性有何不同?

How do software characteristics differ from hardware characteristics ?

- Software is developed or engineered it is not manufactured in the classical sense.
- Software doesn't "**wear out**" but is does **deteriorate**.
- Although the industry is moving toward component-based construction, most software continues to be custom-built.





1.3 What is wrong with the notion that computer software does not need to evolve over time?

中文翻译： “计算机软件不需要随时间演化”这一观点错在哪里?

Why must change ?（What is wrong with the notion that software does not need to evolve over time ?）

- The software must be **adapted** to meet the needs of **new computing environments** or **technology**.
- The software must be **enhanced** to implement **new business requirements**.
- The software must be **extended** to make it work with other **more modern systems** or **databases**.
- The software must be **re-architected** to make it viable within an **evolving computing environment**.

1.4 What factors need to be considered when selecting a process model for a project?

中文翻译： 为项目选择过程模型时需要考虑哪些因素?

What are the possible differences in process models for different projects? (What factors need to be considered when **selecting a process model** for a project?)

- **Overall flow** of activities, actions, and tasks and the **interdependencies** among them.
- **Degree** to which actions and tasks **are defined** within each framework activity.
- **Degree** to which work products **are identified and required**.
- **Manner** which quality assurance activities **are applied**.
- **Manner** in which project tracking and control activities **are applied**.
- **Overall degree of detail and rigor** with which the process is described.
- **Degree** to which the customer and other stakeholders are **involved** with the project.
- **Level of autonomy** given to the software team.
- **Degree** to which team organization and roles **are prescribed**.

1.5 How Polya's problem solving principles describe the essence of engineering practice?

中文翻译： Polya的问题求解原则如何描述工程实践的本质?

How to integrate the engineering project practice into the process framework? (How Polya's problem solving principles describe the essence of engineering practice?)

George Polya 《How to Solve it》 suggests:

1. *Understand the problem* (communication and analysis).
2. *Plan a solution* (modeling and software design).
3. *Carry out the plan* (code generation).
4. *Examine result for accuracy* (testing & quality assurance).

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2. UNIT2 Requirements Engineering — Understanding Requirements（第6章）

2.1 What are the six steps for requirements engineering?

中文翻译： 需求工程的六个步骤是什么?

What are the six steps for requirements engineering?

- **seven** distinct **tasks**: inception, elicitation, elaboration, negotiation, specification, validation, and **management**.

英文答案： Inception, elicitation, elaboration, negotiation, specification, validation.

中文翻译： 启动、获取、细化、协商、规约、验证。

2.2 How do requirement engineers collaborate with stakeholders?

中文翻译： 需求工程师如何与利益相关者协作?

How do requirement engineers collaborate with stakeholders?

- identify **areas of commonality** (i.e., requirements on which all stakeholders agree) and **areas of conflict** or **inconsistency** (i.e., requirements that are desired by one stakeholder but conflict with the needs of another stakeholder).
- use a "**voting**" scheme based on **priority** points.

2.3 Why are nonfunctional requirements important to the requirements engineer?

中文翻译： 为什么非功能需求对需求工程师很重要？ （仅抄上面两段）

Why are NFRs **important** to the requirements engineer?

- NFRs can be described as a set of system attributes, including quality, performance, security, or general constraints, etc.
- often not easy for stakeholders to articulate, yet the software may not be used without including the NFRs.

identify NFRs：**two-phase** approach

- the first phase, **a set of guidelines** and **a list of NFRs** are established …
- the second phase, the team **prioritizes** each NFR by …

✧ 第 1 阶段：开发一套实施**最佳实践的指南**（涉及体系结构风格与模式）和一个**NFR列表**（可用性、可测试性、安全性、可维护性等），列标签表示NFR，行标签表示 SE 指南，**利用关系矩阵比较、评估每一对准则是互补的、重叠的、冲突的还是独立的**

✧ 第 2 阶段：依据一套**决策规则**（决定实施或放弃哪些指南）划分出同类 NFR，并为 NFR 建立优先级

2.4 What work products result from the requirements engineering process?

中文翻译： 需求工程过程会产生哪些工作产品?

What **work products** result from the RE process?

For large systems, the work products may include:

- a **statement** of need and feasibility;
- a **bounded statement** of scope for the system or product;
- a **list** of all stakeholders who participated in requirements elicitation;
- a **description** of the system's technical environment;
- a **list** of requirements and the domain constraints that apply to each;
- a **set** of usage scenarios that provide insight into the use of the system or product under different operating conditions.

Each work product is **reviewed** by all stakeholders.

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3. UNIT3 Requirements Modeling - Building Analysis Model（第7章）

3.1 Which models can be used in requirements modeling, and what role does each type of model play?

中文翻译： 需求建模中可以使用哪些模型？每种模型的作用是什么?

Which models can be used in requirements modeling, and what role does each type of model play?

- **Scenario-based models** depict requirements from the point of view of various system "actors."
- **Class-oriented models** represent OO class and how classes collaborate to achieve system requirements.
- **Behavioral models** depict how the software reacts to internal or external "events."
- **Data models** depict the information domain for the problem.
- **Flow-oriented models** represent functional elements of the system and how they transform data in the system.

3.2 Which UML diagrams are useful in scenario-based modeling?

中文翻译： 哪些UML图在基于场景的建模中有用?

Which **UML diagrams** are useful in scenario-based modeling?

- Using UML to model requirements begins with the creation of scenarios in the form of **use case diagram**s, **activity diagram**s, and **sequence diagram**s.

英文答案： Use case diagrams, activity diagrams, and sequence diagrams.

中文翻译： 用例图、活动图和序列图。

3.3 What questions should be asked to help refine a preliminary use case?

中文翻译： 为帮助细化初步用例，应提出哪些问题?

What questions should be asked to help **refine** a preliminary use case?

Description of **alternative** interactions is **essential** to **completely understand** a function described a use case.

- Can the actor take some **other action** at this point ?
- Is it possible that the actor will encounter some **error condition** at this point ?
- Is it possible that the actor will encounter some **other behavior** at this point (for example: behavior that is invoked by some event outside the actor's control) ?

3.4 What characteristics should be considered when identifying potential classes?

中文翻译： 识别潜在类时应考虑哪些特征？

What characteristics should be considered when identifying (selecting) potential classes?

- **Retained information**. Potential class will be useful during analysis only if information about it must be remembered.
- **Needed services**. Potential class must have a set of identifiable operations that can change the value of its attributes in some way.
- **Multiple attributes**. Focus on "major" information; a class with a single attribute may be better represented as an attribute of another class.
- **Common attributes**. A set of attributes can be defined for the potential class and the attributes apply to all instances of the class.
- **Common operations**. A set of operations can be defined for the potential class and the operations apply to all instances of the class.
- **Essential requirements**. External entities that appear in the problem space and produce or consume information essential to the solution will usually be defined as analysis classes in the model.

3.5 What are the steps needed to build a behavioral model?

中文翻译： 构建行为模型需要哪些步骤？

What are the **steps** needed to **build** a behavioral model?

1. **Evaluate** all use cases to fully understand the sequence of interaction within the system.
2. **Identify** events that drive the interaction sequence and **understand** how these events relate to specific objects.
3. **Create** a sequence diagram for each use case.
4. **Build** a state diagram for the system.
5. **Review** the behavioral model for accuracy and consistency.

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4. UNIT4 Design & Architecture（第8、9章）

4.1 How effective modular design is achieved through functional independence of the individual modules?

中文翻译： 如何通过单个模块的功能独立性实现有效的模块化设计？

How effective modular design is achieved through functional independence of the individual modules?

Design Goal: **high cohesion** & **low coupling**.

independence is achieved by making modules single-minded and preventing excessive interaction with other system elements.

- design should ensure each module addresses a specific subset of requirements and has a simple interface.

independence is a key to good design, and design is the key to software quality. **Independent module** is

- easier to develop, because function can be compartmentalized and interfaces are simplified, and easier to perform parallel implementation of modules.
- easier to maintain & test, because secondary effects caused by modification are limited, error propagation is reduced, and reusable modules are possible.

4.2 What is the principle of information hiding as it applies to software design?

中文翻译： 应用于软件设计的信息隐藏原则是什么?

What is the principle of information hiding as it applies to software design?

- Principle suggests that modules should be " characterized by design decisions that (each) hides from all others ".
- Hiding implies that effective modularity can be **achieved** by defining a set of **independent modules** that communicate with one another only that information necessary to achieve software function.
- Abstraction helps to define entities (**procedural** or **informational**) that make up the software.
- Hiding defines & enforces access constraints to **both procedural detail** within a module **and** any **local data structure** used by the module.

4.3 What is design refactoring?

中文翻译： 什么是设计重构?

What is design refactoring?

- a reorganization technique of changing a system in such a way that simplifies the design (or code) without altering **functionality and the external behavior** yet improves its **internal structure**.

4.4 Why is architecture important?

中文翻译： 为什么架构很重要?

Why is architecture important?

three **key** reasons:

1. provides a representation that **facilitates** communication among all stakeholders.
2. highlights **early design decisions** that will have a **profound impact** on all **SE** work that follows, and plays an important role in the success of the system as **a runnable entity**.
3. constitutes a relatively small model (**a set of abstractions**) of how the system components are structured and **work together**.

4.5 What are the elements that make up a software architectural style?

中文翻译： 组成软件架构风格的元素有哪些?

What are the **elements** that make up the software architectural **style** ?

Each **style** describes a system **category** that encompasses:

1. **a set of components** (e.g., a database, computational modules) that perform a function required by a system,
2. **a set of connectors** that enable "communication, coordination and cooperation" among components,
3. **constraints** that define how components can be integrated to form the system, and
4. **semantic models** that enable a designer to understand the overall properties of a system by analyzing the known properties of its constituent parts.

4.6 What is an archetype?

中文翻译： 什么是原型（archetype）？

What is an archetype ?

The target system architecture is composed of a **relatively small set** of **archetypes**.

Archetype design is **critical** to the design of an architecture.

An **archetype**

- is a **class** or **pattern** that represents **stable** element with a **core abstraction** for the target system.
- may be **instantiated iteratively** many **different ways** based on the **behavior** of the system.
- can be **derived** by examining the **analysis classes** defined as part of the requirements model.

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5. UNIT5 Component-Level Design & UI Design（第10、11章）

5.1 How does the object-oriented view of component-level design differ from the traditional view?

中文翻译： 面向对象的组件级设计视图与传统视图有何不同?

How does the **OO-View** of component-level design differ from the **traditional-view** ?

OO-View: a component contains a set of collaborating classes.

- A class is related to the problem domain and infrastructure domain.
- Each class within a component must be elaborated by defining the data structure of attributes, describing the process flow of operations.
- Specify the message details of all interfaces between the components.

Traditional-View: a module is a functional element of a program.

- three type of modules: control, business, infrastructure.
- All modules requires representations to be created for processing logic, the internal data structures, and an interface that enables the modules to be invoked and data to be passed to it.

A Process-Related View: CBSE

- assembling architecture by **reusing** components or design patterns.

5.2 What are the steps used to complete the component-level design for a software development project?

中文翻译： 完成软件开发项目的组件级设计需要哪些步骤?

What are the **steps** used to complete the **component-level** design for a software development project?

1. Identify **design** classes that correspond to problem domain.
2. Identify **design** classes that correspond to infrastructure domain.
3. Elaborate all **design** classes that are not acquired as reusable components.
4. Identify persistent **data** sources & identify the classes required to **manage** them.
5. Develop & elaborate **behavioral** representations for each class or component.
6. Elaborate **deployment** diagrams to provide additional implementation **detail**.
7. Refactor every **component-level** representation and always consider alternatives.

5.3 What principles should be followed when building any user interface?

中文翻译： 构建任何用户界面时应遵循哪些原则?

What **principles** should be followed when building any **user interface**?

- Place the user in control.
- Reduce the user's memory load.
- Make the interface consistent.

5.4 What framework activities need to be completed in the development process of user interface evolution (or spiral)?

中文翻译： 在用户界面演化（或螺旋）开发过程中需要完成哪些框架活动?

What framework activities need to be completed in the development process of user interface evolution (or spiral) ?

Interface analysis and modeling

- focus on the profile of the users who will interact with the system.

Interface design

- defines a set of interface objects and actions that enable a user to perform all defined tasks in a manner that meets every usability goal defined for the system.

Interface construction

- normally begins with the creation of a prototype that enables usage scenarios to be evaluated.

Interface validation

- focuses on:
  1. The ability of interface to implement every user task correctly.
  2. The degree to which interface is easy to use and easy to learn.
  3. The user's acceptance of the interface as a tool in her work.

5.5 What things you need to do in user experience analysis if you seek to "understand the problem before you attempt to design a solution."?

中文翻译： 如果你想“在尝试设计解决方案之前理解问题”，在用户体验分析中需要做哪些事情?

What things you need to do in UX analysis if you seek to "understand the problem before you attempt to design a solution."?

In the case of **UX** design, understanding the problem **means understanding** :

- the people (end users) who will interact with the system through the interface,
- the tasks that end users must perform to do their work,
- the content that is presented as part of the interface, and
- the environment in which these tasks will be conducted.

5.6 How to evaluate user interface design before building a working computer prototype?

中文翻译： 在构建可工作的计算机原型之前如何评估用户界面设计?

How to **evaluate UI design** before building a working computer prototype ?

If **design model** of **UI** has been created, some **evaluation criteria** can be applied during early design reviews:

1. length & complexity of the specification (**design model**)
   ⟹ user's learning effort
2. number of user tasks & average number of user actions per task
   ⟹ system's (interaction time & overall efficiency)
3. number of tasks, actions, and system states
   ⟹ user's memory load
4. Interface style, help facilities & error handling protocol
   ⟹ interface's complexity & user's acceptance degree.

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6. UNIT6 Software Testing（第15、16章）

6.1 How does software verification differ from validation?

中文翻译： 软件验证与确认有何不同?

How does the software **verification** differ from **validation** ?

Verification : a **set of tasks** that ensure the software correctly **implements** a specific function.

Are we **building** the product **right** ?

Validation : a **set of tasks** that ensure the software has been built is **traceable** to requirements.

Are we building the **right product** ?

Testing provide the **last bastion** from which quality can be assessed and, **more pragmatically**, errors can be uncovered.

6.2 What is the difference between black-box testing and white-box testing?

中文翻译： 黑盒测试与白盒测试的区别是什么?

What is the difference between black-box testing and white-box testing?

Black-box testing involves testing the functionality of a software component without knowing the details of its internal logic.

White-box testing involves testing the independent logic paths with full implementation knowledge.

6.3 What is object-oriented unit testing?

中文翻译： 什么是面向对象的单元测试?

What is object-oriented unit testing?

OO **class** testing is equivalent to **unit** testing of conventional software.

- Conventional unit testing focus on the **algorithmic detail** of a module and the **data** that flow across the module interface.
- OO class testing is driven by the **operations** encapsulated by the class and the **state behavior** of the class.
- The focus throughout is on **designing and testing appropriate sequences of operations** to **exercise all class states**.
- It makes little sense to **test operations or algorithms individually** for classes.

6.4 What are the attributes of a good software test?

中文翻译： 好的软件测试具有哪些属性?

What are the attributes of a good software test?

- Has a high probability of finding an error
- Not redundant
- Should be "best of breed": the capable of uncovering a whole class of errors
- Should not be too simple or too complex

6.5 Why is regression testing an important part of any integration testing procedure?

中文翻译： 为什么回归测试是任何集成测试过程中的重要部分?

Why is regression testing an important part of any integration testing procedure ?

- As software is corrected, some aspect of software configuration is changed. Goal of integration testing is to **make sure** that **independent modules that work correctly** on their own do not interfere with one another when added to the same program.
- Regression testing helps to ensure that **do not introduce unintended behavior** or **additional errors**.

6.6 What are the key differences between validation testing goals and acceptance testing goals?

中文翻译： 确认测试目标与验收测试目标之间的主要区别是什么？ （仅抄上面两段）

What are the key differences between validation testing goals and acceptance testing goals ?

- In validation testing, the test team seeks to ensure that each software function or performance characteristic conforms to its specification.
- In acceptance testing, the test team needs to ensure that the software works correctly for the intended user in his or her normal work environment.

acceptance testing: Alpha/Beta testing, focus on customer usage

- α 测试（受控）：由代表性最终用户，在开发者现场进行，开发者在后面观看，记录使用中的错误和问题
- β 测试（客户验收）：按合同交付软件时，用户场所进行，开发者不在场（不受控），用户记录问题并提交

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7. UNIT7 Software Configuration Management（第17章）

7.1 What information will make up the software configuration?

中文翻译： 哪些信息构成软件配置?

What information will make up the software configuration?



- Software configuration are work products that created by engineers during the software process.

  computer programs (source code and executables), documentation (technical and user), data (internal and external to programs)

7.2 What are the steps in the change control process of a project?

中文翻译： 项目变更控制过程的步骤是什么?

What are the **steps** in the change control process?

1. **evaluate** a change request, and **submit** a change report to **CCA** (change control authority). The **CCA** makes the final determination as to the **status** and **priority** of the change.
2. An **ECO** (engineering change order) is generated for each **approved change**.
3. Change SCIs are **checked out** from the project database subject by its access control **parameters**.
4. The **modified SCIs** is subjected to **SQA** procedures and **checked into** the project database.
5. **Version control** procedures are followed to produce the **next version** of the software. Adopting **Synchronization control** ensures that **parallel changes** made by different people do **not overwrite** with each other.

7.3 What is a software configuration audit?

中文翻译： 什么是软件配置审计?

What is a software configuration audit?

- **Technical Review**: focuses on the technical correctness of the configuration object that has been modified. Reviewers assess the **SCI** to determine consistency with other **SCI**s, omissions, or potential side effects.

- **Configuration audit**: is a supplement to technical review.

  Once a change has been made to a **SCI** and an **FTR** has been conducted, the **software quality team** conducts its own review to ensure that software process and standards have been followed, including updating all affected documents and any other affected **SCI**s.

7.4 What is impact management?

中文翻译： 什么是影响管理?

What is impact management ?

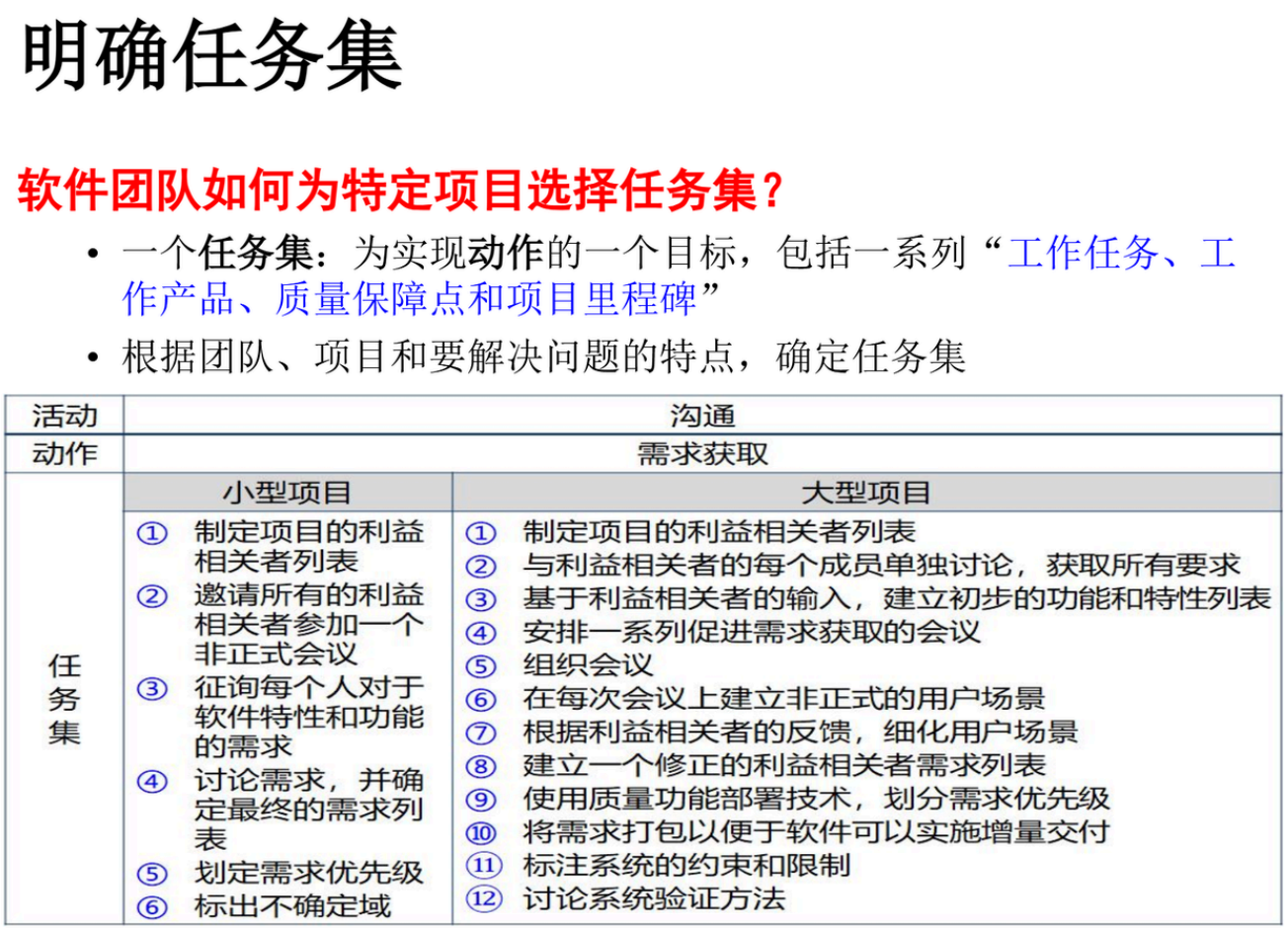
A web of software work product interdependencies must be considered when a change is made, it is necessary to properly understand these interdependencies and control their effects on other SCIs.

8. UNIT8 Software Process & Process Model（第2、3、4章）

关注题目 1、3、4、6

8.1 How does software team choose the task set for a particular project?

中文翻译： 软件团队如何为特定项目选择任务集？



英文答案：

A software team defines and selects a specific task set **based on the characteristics of the team, the project, and the problem to be solved.**

A **task set** is a collection of engineering work designed to achieve the goal of a specific action. It includes a series of **work tasks, work products, quality assurance points, and project milestones.**

Task Set Selection for a Small Project

For a small project, the team selects a streamlined and flexible task set consisting of 6 core tasks:

1. Develop a list of project stakeholders.
2. Invite all stakeholders to an informal meeting.
3. Solicit each person's requirements for software features and functions.
4. Discuss requirements and determine the final requirements list.
5. Prioritize the requirements.
6. Note areas of uncertainty.

Task Set Selection for a Large Project

For a large project, where the problem and environment are much more complex, the team chooses a more rigorous and systematic task set expanded into 12 detailed tasks:

1. Develop a list of project stakeholders.
2. Interview each stakeholder individually to gather all requirements.
3. Build an initial list of functions and features based on stakeholder input.
4. Schedule a series of facilitation meetings for requirements elicitation.
5. Organize and conduct the meetings.
6. Develop informal user scenarios at each meeting.
7. Refine user scenarios based on stakeholder feedback.
8. Build a revised stakeholder requirements list.
9. Use Quality Function Deployment (QFD) techniques to prioritize requirements.
10. Package requirements so that the software can be implemented in incremental deliveries.
11. Document system constraints and restrictions.
12. Discuss methods for system validation.

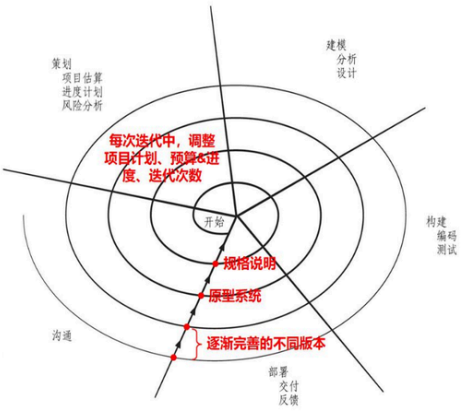
8.2 Why the spiral model is the best approach to software development in a modern context?

中文翻译： 为什么螺旋模型是现代背景下软件开发的最佳方法?

螺旋过程模型

Why the spiral model is the best approach to software development in a modern context?

- timelines for the development of modern software are getting shorter and shorter
- customers are becoming more diverse and making the understanding of requirements even harder
- changes to requirements are becoming even more common before delivery
- need a way to provide incremental or evolutionary delivery to better adapt to uncertainty, allows the delivery of partial solutions in an orderly and planned manner
- most importantly, reflects what really happens when complex systems are built



8.3 What are the key issues stressed by an agile philosophy?

中文翻译： 敏捷哲学强调的关键问题是什么?

agile philosophy

What are the key issues stressed by an agile philosophy?

1. Emphasis on rapid delivery of software that satisfies the customer
2. Recognition that change represents opportunity
3. Communication and collaboration between team members and customers
4. The importance of self-organizing teams

8.4 What are the tradeoffs proposed by the "Agile Manifesto"?

中文翻译： “敏捷宣言”提出了哪些权衡?

《敏捷宣言》提出了哪些折衷方案 ？

1. 个人之间的交流**胜过**开发过程和工具
2. 可运行软件**胜过**宽泛文档
3. 客户合作**胜过**合同谈判
4. 良好响应变更**胜过**按部就班地遵循计划

英文答案：

Individuals and interactions over processes and tools.

Working software over comprehensive documentation.

Customer collaboration over contract negotiation.

Responding to change over following a plan.

8.5 What are the three questions that should be answered by each team member at the daily Scrum meeting?

中文翻译： 每日Scrum会议中每个团队成员应回答的三个问题是什么?

What are the three questions that should be answered by each team member at the daily Scrum meeting ?

- What did you do since the last team meeting?
- What obstacles are you encountering?
- What do you plan to accomplish by the next team meeting?





8.6 Why should requirements engineering be an iterative process?

中文翻译： 为什么需求工程应该是一个迭代过程?

Why should requirements engineering be an iterative process?

- Two facts of requirement definition: ① It is impossible for stakeholders to describe an entire system before seeing the working software; ② it is difficult for stakeholders to describe quality requirements needed for the software before seeing it in action as a prototype.
- Developers must recognize that requirements will be added and refined as the increments are created.
- Using prototypes may increase the volatility of the requirements if stakeholders are not focused on getting things right the first time.

8.7 Why is it important to test a prototype using the stakeholders?

中文翻译： 为什么让利益相关者测试原型很重要?

Why is it important to test a protype using the stakeholders ?

- getting feedback through testing the prototype, and risk assessment based on "user satisfaction" and "whether the budget is exceeded and delivery is delayed", making a decision on whether to create the next prototype.
- Using stakeholders is essential to reduce the risk of developing the wrong product.
- Having developers do all the testing is not wise because they are not likely to be the representative of the intended user population.
- It's important to have the right mix of users (e.g., novice, typical, and advanced) to give you meaningful feedback on the prototype.

8.8 What are the types of maintenance activities?

中文翻译： 维护活动的类型有哪些?

What are the types of maintenance activities?

- Corrective maintenance - reactive modification to **repair problems** discovered after delivery.
- Adaptive maintenance - reactive modification to **keep** software **usable in a changing context** after delivery.
- Perfective maintenance - proactive modification to **provide new** user features, **better** program code structure, or **improved** documentation after delivery.
- Preventive maintenance – proactive modification to **correct product faults** before discovery by users after delivery.

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9. UNIT9 Project Management & Software Plan（第5、19、20章）

9.1 What are four P's of effective project management?

中文翻译： 有效项目管理的四个P是什么？

What are four P's of effective project management?

- **People** — the most important element of a successful project.
- **Product** — the software to be built.
- **Process** — the set of framework activities and software engineering tasks to get the job done.
- **Project** — all work required to make the product a reality.

英文答案： People, Product, Process, Project.

中文翻译： 人员、产品、过程、项目。

9.2 What are the characteristics of software engineers should possess?

中文翻译： 软件工程师应具备哪些特征？

What are the characteristics of software engineers should posess?

- Sense of individual responsibility
- Has an acute awareness of the needs of other team members
- Brutal honesty
- Resilience under pressure
- Heightened sense of awareness
- Attention to detail
- Takes a pragmatic approach to software engineering

9.3 What are the key attributes of an effective software teams?

中文翻译： 高效软件团队的关键属性是什么？

What are the key attributes of an effective software teams?

- Sense of purpose.
- Sense of involvement.
- Sense of trust.
- Sense of improvement.
- Diversity of team member skill sets.

9.4 What are the environment characteristics that can be considered toxic to software teams? How to avoid it?

中文翻译： 哪些环境特征对软件团队是有毒的？如何避免？

What are the environment characteristics that can be considered toxic to software teams? How to avoid it?

Toxic factors	Avoiding Solution
1. **Frenzied work atmosphere** where team members waste energy and lose focus. 2. **High frustration** that causes friction among team members 3. **Fragmented or poorly coordinated process model** that becomes a roadblock to accomplishment. 4. **Unclear definition of team roles** resulting in a lack of accountability and resultant finger-pointing. 5. **Continuous and repeated exposure to failure that** leads to a loss of confidence and poor morale.	✧ Be certain that the team has access to all information required to do the job. ✧ Major goals and objectives should not be modified unless absolutely necessary. ✧ Team should be given as much decision-making responsibility as possible. ✧ Allow the team to select the process model. ✧ Team should establish its own mechanisms for accountability. ✧ Establish team-based techniques for feedback and problem solving.
6. Differing and incompatible team member human traits.	

9.5 How can agile teams avoid toxicity that affects the context of project?

中文翻译： 敏捷团队如何避免影响项目环境的有毒因素？

How can a agile team avoid toxicity that affects the context of project?

- ✧ encourages customer satisfaction and early incremental delivery of software;
- ✧ small, highly motivated project teams;
- ✧ informal methods;
- ✧ minimal SE work products; and
- ✧ overall development simplicity.

9.6 How is software scope defined?

中文翻译： 如何定义软件范围?

How is software scope defined?

Statement of software scope must be **bounded**, **unambiguous** and **understandable** at the management and technical levels.

Context: Defining how the software to be built fits into a larger systems, product, or business context and the constraints imposed by the context.

Information objectives: Identifying what customer expects to see as visible output and what input objects are required to produce them.

Function and performance: Determining the function needed transform input to output and any special performance characteristics.

Attempting to bound all information quantitatively, when possible, and descriptively, when numbers cannot be used.

可定量界定：并发用户数、邮件列表长度、处理时间、最大响应时间等性能需求，产品成本要求、内存大小等约束条件

不可定量界定：期望算法易理解，并采用 Java 实现

9.7 What are the questions that need to be answered using the W^5H^2 ?

中文翻译： 使用W^5H^2方法需要回答哪些问题?

What are the questions that need to be answered using W^5H^2 ?

By posing a set of questions that lead to a definition of key project characteristics and the resultant project plan, describe project objectives, milestones and schedules, responsibilities, management and technical approaches, and required resources.

- Why is the system being developed?
- What will be done?
- When will it be done?
- Who is responsible for a function?
- Where are they located organizationally?
- How will the job be done technically and managerially?
- How much of each resource is needed?